

Reliable voltage-induced magnetization switching for voltage-controlled MRAM

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Voltage-induced magnetization switching is a promising approach for realizing ultra-low-power spintronic devices. In particular, voltage-controlled MRAM based on the voltage-controlled magnetic anisotropy (VCMA) effect has attracted considerable attention as a highly energy-efficient alternative to current-induced switching. However, conventional voltage-induced dynamic switching requires extremely precise control of the voltage pulse width, posing challenges to reliable operation and large-scale integration. Here, we propose a novel voltage-induced magnetization switching scheme based on the VCMA effect and an artificial antiferromagnetic multilayer, termed “voltage-driven static magnetization switching (VCMA-SMS)”. This method enables energy-efficient switching for voltage-controlled MRAM and achieves robust bidirectional switching over a wide voltage pulse width range from tens of nanoseconds to quasi-static states, providing high robustness against pulse-width variations. The proposed approach opens a pathway toward ultra-low-power spintronic memory applications.

Index Terms—Spintronics, Perpendicular magnetic anisotropy, Magnetic multilayers, Magnetic devices

I. INTRODUCTION

THE rapid development of artificial intelligence (AI) and Internet-of-Things (IoT) technologies is driving a growing demand for energy-efficient information processing. Nonvolatile memory devices are essential for reducing power consumption, and magnetoresistive random-access memory (MRAM) has attracted considerable attention due to its high-speed operation and high write endurance [1]. However, spin-transfer-torque (STT)-MRAM requires large write currents, resulting in substantial Joule energy loss.

Voltage-induced magnetization switching based on the voltage-controlled magnetic anisotropy (VCMA) effect is attracting attention as a promising alternative that enables switching without the need for large currents [2], [3]. The VCMA effect typically reduces the perpendicular magnetic anisotropy (PMA) of a ferromagnet by applying a positive voltage. Conventional VCMA-driven methods achieve magnetization switching via precessional dynamics induced by sub-nanosecond voltage pulses [4], [5]. However, such dynamic switching requires extremely precise pulse-width control, and even slight deviations lead to high write error rates, posing a serious challenge for large-scale integration.

In this work, we aim to address this fundamental limitation by introducing an artificial antiferromagnetic (AFM) multilayer composed of ultrathin ferromagnetic layers. By exploiting the interplay between VCMA-induced anisotropy modulation and interlayer exchange interaction, the magnetic energy landscape can be continuously tuned by voltage, leading to an alternative switching method called voltage-driven static magnetization switching (VCMA-SMS) [6]. This method enables robust switching over a wide range of voltage pulse conditions and provides an energy-efficient pathway for next-generation spintronic devices.

II. VOLTAGE CONTROL OF SWITCHING FIELD IN ARTIFICIAL AFM

The multilayer device consists of a ferromagnet/non-magnet/ferromagnet trilayer structure exhibiting interlayer exchange coupling (IEC), enabling the formation of an artificial antiferromagnet (AFM). The magnetization configuration switches between antiparallel (AP) and parallel (P) states depending on the exchange coupling and the effective magnetic field.

In this system, voltage control of the switching field can be expected through the modulation of the effective exchange field via the VCMA effect. Specifically, the application of a voltage modifies the PMA of one ferromagnetic layer, which in turn changes the switching field of the artificial AFM without directly modifying the intrinsic IEC strength.

III. VCMA DRIVEN STATIC MAGNETIZATION SWITCHING (VCMA-SMS)

By utilizing the voltage-controlled switching field in the artificial AFM, VCMA-SMS between AP and P states can be achieved under a fixed external magnetic field (see Fig. 1). Unlike conventional dynamic switching, this process does not rely on magnetization precession and corresponds to transitions between energy minima.

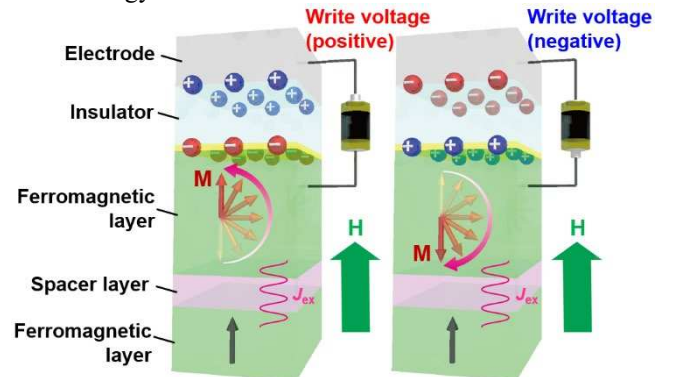


Fig. 1. Concept of VCMA-SMS in artificial AFM.

To demonstrate VCMA-SMS, we fabricated Co/Ru/Co-based artificial AFM multilayers and investigated their magnetic properties using polar magneto-optical Kerr effect (p-MOKE) measurements. The switching field of the artificial AFM is strongly dependent on the Ru spacer thickness, showing a maximum value around 0.4 nm, which is consistent with the oscillatory nature of IEC [7].

By introducing ultrathin heavy metal layers at the interface, the PMA of the top ferromagnetic layer is tailored [8], [9], which directly affects the effective switching field of artificial AFM. When a voltage is applied, the switching field is linearly modulated with an efficiency of several tens of mT/V (see Fig. 2(a)). Here, a clear correlation is observed: the switching field increases (decreases) with increasing PMA for VCMA materials exhibiting negative (positive) VCMA [10], demonstrating a strong coupling between interfacial anisotropy and IEC.

The strong VCMA-IEC coupling enables clear voltage-induced switching between AP and P states under a fixed magnetic field (see Fig. 2(b)). The voltage-induced switching is reproducible and can be driven by both DC and pulse voltages.

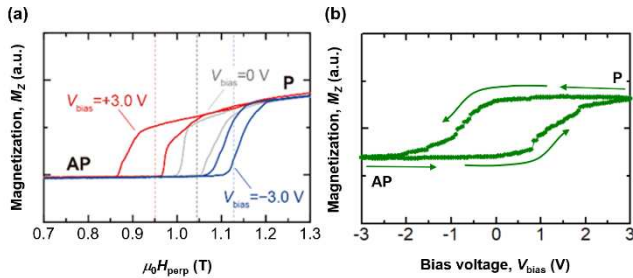


Fig. 2. (a) Bias-voltage-dependent p-MOKE magnetization curves in the vicinity of the switching field [6]. (b) Bias voltage dependence of the p-MOKE signal under fixed perpendicular magnetic field.

A key feature of this VCMA-SMS is its robustness to pulse-width variations. In contrast to conventional VCMA-driven dynamic switching that requires sub-nanosecond precision, stable switching is achieved over a wide pulse width range from tens of nanoseconds to DC conditions. Experimental measurements have demonstrated reliable switching down to approximately 50 ns, and theoretical considerations suggest that sub-nanosecond switching is achievable in nanoscale single-domain devices. Furthermore, bidirectional switching is realized simply by reversing the voltage polarity. This eliminates the need for pre-reading the initial magnetic state, simplifies circuit design, and improves operation speed.

IV. VCMA-SMS WITH HEAVY METALS

The present method introduces a fundamentally different design principle from conventional switching schemes. Since VCMA-SMS does not rely on magnetization precession, large Gilbert damping is not detrimental; in fact, it may even enhance switching speed.

This feature allows the use of materials with strong spin-orbit coupling, such as heavy-metal-based ferromagnets, which

are usually considered unsuitable for dynamic switching. Consequently, VCMA-SMS significantly expands the range of material systems applicable to spintronic device design.

V. CONCLUSION

We have demonstrated a novel voltage-induced static magnetization switching mechanism based on artificial AFM multilayers. By exploiting voltage control of the effective switching field of the artificial AFM multilayers, we achieved reliable magnetization switching over a wide range of pulse conditions. Although some challenges remain, such as reducing the required external magnetic field, further enhancing the VCMA efficiency for practical device applications, and developing MTJ structures suitable for device integration, this method overcomes the major limitations in conventional VCMA switching and provides a promising pathway toward large-scale integration and ultra-low-power consumption of spintronic devices based on the VCMA effect.

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